Transfer: energy and work

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Choose and justify energy methods in new physical situations.

SUCCESS CRITERIA - I can...

1. Choose the correct energy terms and units.
2. Show a consistent numerical balance.
3. Explain what your calculation treats as true and when it stops working.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “work” and “power” to explain your answer. Sentence starters are optional.

work: energy transferred by a force along a displacement

power: rate of energy transfer

energy budget: an account of all energy stored and transferred

01 Warm-up: bring back an earlier idea

At fixed mass, speed becomes **1/3** of its original value. **Compare kinetic energy** and the energy-equivalent TNT amount.

Word help: kinetic energy = energy associated with motion

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $W_{net} = \Delta K$; $K = \frac{1}{2}mv^2$; $U_g = mgh$; $U_s = \frac{1}{2}kx^2$
$P = W/\Delta t$.
g = **10 N/kg** where needed. Select relationships for the system and facts stated here.

Worked reasoning: A **2 kg** cart drops **2 m** with g =**10**, then loses **10 J** to friction. From rest, its final kinetic energy is **40**-**10**=**30 J**, so speed √**30**≈**5.48 m/s**.

Your turn: Draw a complete energy account and a sketch identifying where friction acts.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Initial | **Transfers** | **Final**

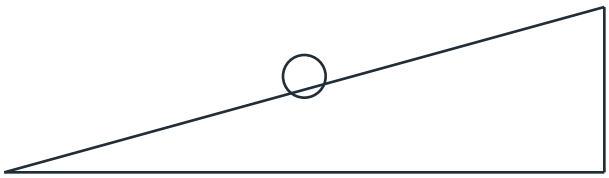
Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A cart coasts down a frictionless ramp from rest. **Explain which energy decreases** and which increases. **Mark the vertical height** in the sketch.

Sentence starter: I think ... . I can point to ..., which shows ..., so ... .



The marker represents the object on the sloping surface.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

A **2 kg** cart drops **2 m**.
1. Use g=**10**.
2. **Find initial gravitational energy** and bottom kinetic energy if losses are negligible.
3. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Word help: negligible = small enough to leave out of this model

Sentence starter: Both ... . They differ in ... because ... .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

05 Explain your reasoning

A motor supplies **100 J** while the load gains **70 J** mechanical energy. **Explain where the other 30 J** might go and whether energy was destroyed.

Sentence starter: I think ... . I can point to ..., which shows ..., so ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

An ideal system has total mechanical energy **24 J**. At a point where U=**4 J**, find K.
Show the energy changes and transfers; **label units**.

Word help: mechanical energy = total kinetic and potential energy in a model

Sentence starter: I found ... [unit]. This value means ... in this situation.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

07 Exit check

A device transfers **60 J** in **5 s**. **Find its average power**. It then transfers the same **60 J** in **6 s**. **Complete the comparison table** and **describe what changes**.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Run	Energy (J)	Time (s)	Power (W)
A	60	5	____
B	60	6	____

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Return to the worked example in problem **2**.
1. Change one feature so the same cart arrives with **20 J** kinetic energy.
2. **Show two different valid designs**.
3. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Sentence starter: Both ... . They differ in ... because ... .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3